A Lipschitz Resolution of the Missing L^1 Bound for Variable-Step Mollifiers

FA/Banach automatic math run

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Source And Target

The source is Michael Hintermuller, Kostas Papafitsoros, and Carlos N. Rautenberg, *Variable step mollifiers and applications*, arXiv:1910.02751, published in *Integral Equations and Operator Theory* 92 (2020), article 53. For an open bounded set $\Omega \subset \mathbb{R}^N$, a non-negative compactly supported mollifier ρ on the unit ball, and a variable step η, the source studies

$$T_n f(x) = C_n(x) \int_\Omega \rho\left(\frac{x-y}{\eta(x)/n}\right) f(y)\, dy, \qquad C_n(x) = M_\rho \frac{n^N}{\eta(x)^N}.$$

In Section 5, after proving the uniform L^1 estimate under the quadratic boundary condition

$$\kappa \,\mathrm{dist}(x, \partial\Omega)^2 \leq \eta(x) \leq \mathrm{dist}(x, \partial\Omega)^2,$$

the source asks whether the estimate remains true without imposing a condition of that type.

ending the proof. ∎

Note that by squaring (5.6) and rearranging the constants we get the existence of a $C^\infty(\Omega)$ function $\eta := \tilde{\eta}^2$ (not to be confused with the initial η used in the proof above) that satisfies

$$\kappa\sigma(x)^2 \leq \eta(x) \leq \sigma(x)^2, \quad \text{for every } x \in \Omega, \tag{5.11}$$

for some $0 < \kappa < 1$. Observe that the operator T can be also defined using this η and everything proven so far still holds for this case as well. In the next proposition we show that provided η satisfies the condition (5.11) then (5.1) holds and hence $T \in \mathscr{L}(L^1(\Omega))$. In fact we will show that the resulting sequence $(\|T_n\|_{\mathscr{L}(L^1(\Omega))})_{n\in\mathbb{N}}$ is uniformly bounded.

Theorem 5.3. *Let $\Omega \subseteq \mathbb{R}^N$ open and $\eta \in C^\infty(\Omega)$ with the property that there exists $\kappa > 0$ such that (5.11) holds. Then*

$$\sup_{n\in\mathbb{N}} \sup_{y\in\Omega} \int_\Omega C_n(x)\rho\left(\frac{x-y}{\frac{1}{n}\eta(x)}\right) dx < \infty, \tag{5.12}$$

and hence, T and T_n for $n \in \mathbb{N}$ belong to $\mathscr{L}(L^1(\Omega))$.

Proof. Note that it suffices to show

$$\sup_{n\in\mathbb{N}} \sup_{y\in\Omega} \int_\Omega \frac{n^N}{\eta(x)^N} \mathcal{X}_{B_{\frac{\eta(x)}{n}}(x)}(y)\, dx < \infty. \tag{5.13}$$

Note moreover that the map $x \mapsto C(x)\rho\left(\frac{x-y}{\eta(x)}\right)$ is smooth of compact support. Thus it suffices to show (5.13) only for y's that belong to a small neighbourhood of the boundary. For that, we will work with $y \in \Omega$ such that

$$\sigma(y) < \frac{1}{8}. \tag{5.14}$$

In fact we may assume, via a rescaling argument, that (5.14) holds for every $y \in \Omega$. Using the layer cake representation formula and Fubini's theorem, we get for every $n \in \mathbb{N}$ and $y \in \Omega$

$$\begin{aligned}
\int_\Omega \frac{n^N}{\eta(x)^N} \mathcal{X}_{B_{\frac{\eta(x)}{n}}(x)}(y)\, dx &= \int_\Omega \left(\int_0^\infty \mathcal{X}_{\left\{s\in\Omega:\, \frac{n^N}{\eta(s)^N} \geq t\right\}}(x) dt \right) \mathcal{X}_{B_{\frac{\eta(x)}{n}}(x)}(y)\, dx \\
&= \int_0^\infty \left(\int_\Omega \mathcal{X}_{\left\{s\in\Omega:\, \frac{n^N}{\eta(s)^N} \geq t\right\}}(x) \mathcal{X}_{B_{\frac{\eta(x)}{n}}(x)}(y)\, dx \right) dt \\
&= \int_0^\infty \mathcal{L}^N \Bigg(\underbrace{\left\{ x \in \Omega : \frac{\eta(x)}{n} \leq \frac{1}{\sqrt[N]{t}} \text{ and } y \in B_{\frac{\eta(x)}{n}}(x) \right\}} \Bigg) dt. \qquad (5.15)
\end{aligned}$$

$$\|T_n\|_{\mathscr{L}(L^1(\Omega))} = \sup_{y\in\Omega}\int_\Omega C_n(x)\rho\left(\frac{x-y}{\frac{1}{n}\eta(x)}\right)dx \le \sup_{y\in\Omega} M_\rho \int_\Omega \frac{n^N}{\eta(x)^N}\mathcal{X}_{B_{\frac{\eta(x)}{n}}(x)}(y)\,dx$$
$$\le M_\rho \sup_{y\in\Omega}\left(\omega_N + \omega_N N \ln\left(\frac{1}{\kappa}\frac{n}{n-|y-\xi_y|}\right)\right)$$
$$\le M_\rho\omega_N\left(1+N\ln\left(\frac{1}{\kappa}\frac{n}{n-\frac{1}{8}}\right)\right),$$

something that implies that

$$\limsup_{n\to\infty}\|T_n\|_{\mathscr{L}(L^1(\Omega))} \le M_\rho\omega_N\left(1+N\ln\left(\frac{1}{\kappa}\right)\right). \tag{5.23}$$

Note: It still remains unanswered if the estimate (5.12) (or (5.13)) holds without imposing any condition of the type (5.11). We leave that as an open question.

For the following results that deal with L^1 integrability, we will always assume that the operator T is defined via a function η that satisfies (5.11). We next state the corresponding approximation result in $L^1(\Omega)$:

Proposition 5.4. *Let $f \in L^1(\Omega)$. Then $T_n f \to f$ in $L^1(\Omega)$ as $n \to \infty$.*

Proof. The proof follows exactly the same steps as the one of Proposition 3.2, by taking advantage of the fact that the sequence $(\|T_n\|_{\mathscr{L}(L^1(\Omega))})_{n\in\mathbb{N}}$ is uniformly bounded. □

In view now of Proposition 5.4 and from the lower semicontinuity of the operator norm with

The exact estimate to prove is

$$\sup_{n\in\mathbb{N}}\sup_{y\in\Omega}\int_\Omega \frac{n^N}{\eta(x)^N}\mathbf{1}_{B_{\eta(x)/n}(x)}(y)\,dx < \infty. \tag{1}$$

This implies the source's estimate (5.12), because $0 \le \rho \le 1$ and $C_n(x) = M_\rho n^N/\eta(x)^N$.

Idea Of The Proof

The quadratic lower bound in the source proof is used to prevent the moving ball centered at x from seeing a fixed point y across too many boundary scales. A simpler mechanism is available. If η has Lipschitz constant $L < 1$, then whenever

$$|x-y| < \frac{\eta(x)}{n},$$

the values $\eta(x)$ and $\eta(y)$ are automatically comparable:

$$\frac{n}{n+L}\eta(y) < \eta(x) < \frac{n}{n-L}\eta(y).$$

Thus every relevant x lies in one ordinary ball around y of radius comparable to $\eta(y)/n$, and the integrand is bounded by a constant times $n^N/\eta(y)^N$. Multiplying that height by that volume gives a constant independent of y, n, and the boundary geometry.

Theorem 1 (Lipschitz variable steps satisfy the L^1 kernel bound)**.** *Let $\Omega \subset \mathbb{R}^N$ be open and let $\eta : \Omega \to (0,\infty)$ be an L-Lipschitz function with $0 \le L < 1$. Suppose also that*

$$B_{\eta(x)}(x) \subset \Omega \qquad (x \in \Omega).$$

Then

$$\sup_{n\in\mathbb{N}}\sup_{y\in\Omega}\int_\Omega \frac{n^N}{\eta(x)^N}\mathbf{1}_{B_{\eta(x)/n}(x)}(y)\,dx \le \omega_N\left(\frac{1+L}{1-L}\right)^N,$$

where ω_N is the volume of the unit ball in $\mathbb{R}^N$. Consequently, for every source mollifier ρ,

$$\sup_{n\in\mathbb{N}}\sup_{y\in\Omega}\int_\Omega C_n(x)\rho\left(\frac{x-y}{\eta(x)/n}\right)\,dx \le M_\rho\omega_N\left(\frac{1+L}{1-L}\right)^N.$$

In particular T and all T_n are bounded on $L^1(\Omega)$, uniformly in n.

Proof. Fix $n\in\mathbb{N}$ and $y\in\Omega$, and set

$$A_{n,y} := \{x\in\Omega:\ |x-y|<\eta(x)/n\}.$$

If $x\in A_{n,y}$, then the Lipschitz condition gives

$$\eta(x)-\eta(y)\le L|x-y| < \frac{L}{n}\eta(x),$$

and therefore

$$\eta(x) < \frac{n}{n-L}\eta(y). \tag{2}$$

Similarly,

$$\eta(y)-\eta(x)\le L|x-y| < \frac{L}{n}\eta(x),$$

so

$$\eta(x) > \frac{n}{n+L}\eta(y). \tag{3}$$

Equation (2) also gives

$$A_{n,y}\subset B_{\eta(y)/(n-L)}(y).$$

Using (3) on $A_{n,y}$, we obtain

$$\begin{aligned}\int_\Omega \frac{n^N}{\eta(x)^N}\mathbf{1}_{B_{\eta(x)/n}(x)}(y)\,dx &= \int_{A_{n,y}}\frac{n^N}{\eta(x)^N}\,dx\\ &\le \frac{(n+L)^N}{\eta(y)^N}\left|B_{\eta(y)/(n-L)}(y)\right|\\ &\le \omega_N\left(\frac{n+L}{n-L}\right)^N\\ &\le \omega_N\left(\frac{1+L}{1-L}\right)^N.\end{aligned}$$

This proves (1). The estimate with ρ follows from $0\le\rho\le 1$ and from the formula for C_n. The L^1-boundedness then follows exactly as in Proposition 5.1 of the source by Tonelli's theorem. □

Corollary 1 (No quadratic lower bound is needed for Whitney steps)**.** *In the setting of the source with $\Theta=\partial\Omega$, choose the Whitney step η so that $\|\nabla\eta\|_\infty\le L<1$. Then the source estimates (5.12) and (5.13) hold, and the L^1, $W^{1,1}$, and the $W^{1,1}$ conclusions, and the unmodified* BV *boundedness/weak-star approximation conclusions in Section 5 that used (5.11) remain valid with this Lipschitz step in place of the quadratic step.*

Proof. The source's Whitney construction already permits a harmless rescaling of η, and the proof of its Theorem 2.1 explicitly arranges a uniform gradient bound below 1. Since $\eta = 0$ on $\partial\Omega$, the bound $\|\nabla\eta\|_\infty < 1$ implies $\eta(x) < \operatorname{dist}(x, \partial\Omega)$, hence the balls $B_{\eta(x)}(x)$ lie inside Ω. Theorem 1 gives the missing uniform L^1 kernel estimate. The later L^1-convergence proof, the $W^{1,1}$ proof, and the unmodified BV weak-star proof in the source use the quadratic condition only through that uniform L^1 bound and through boundedness of $\nabla\eta$, both of which are now available. □

Remark 1 (Sharpness of the strict Lipschitz threshold)**.** *The strict inequality $L < 1$ is not cosmetic. The source's one-dimensional example $\Omega = (0, 1)$, $\eta(x) = \operatorname{dist}(x, \partial\Omega)$, has boundary slope 1 and gives an unbounded L^1 operator. On the half-line model $\eta(x) = Lx$, the $n = 1$ kernel mass is bounded for $L < 1$ and grows like $\log(1/y)$ at $L = 1$.*

Verification Notes

The proof is analytic. The script `code/verify_lipschitz_kernel_bound.py` checks the exact one-dimensional linear-step formula

$$\int_0^1 \frac{n}{Lx} \mathbf{1}_{|x-y|<Lx/n}\, dx$$

for several $L < 1$, compares it with the theorem's bound, and illustrates the logarithmic growth at $L = 1$. This is a sanity check for the geometry of the estimate, not a substitute for the proof above.

Novelty Check And Limitations

Cheap run indexes were searched for arXiv:1910.02751, variable-step mollifier keywords, and the strings `eta_bounded_quadratic`, `integrability_full`, and `integrability_suff`. No prior run packet or attempt on this exact question was found. A bounded web search for the same exact strings and for close variants of "Variable Step Mollifiers" with "open question", "quadratic", and "L1" found the source article and no later answer to this estimate.

The theorem does not claim that every possible step $\eta < \operatorname{dist}(\cdot, \partial\Omega)$ works. The source itself gives a negative example at the borderline boundary slope 1. It also does not reprove the strict BV convergence statement for the source's separate modified operators $\tilde{T}_n$. The result proves that the quadratic comparability hypothesis is unnecessary for the intended Whitney-style smooth steps: a strict Lipschitz contraction of the distance scale is enough.

Human Review Recommendation

Review as a claimed full solution of the Section 5 open estimate in the Whitney-step setting. The central proof check is whether the set $|x - y| < \eta(x)/n$ indeed forces the two-sided comparison $\eta(x) \simeq \eta(y)$ with constants uniform in n; the remaining volume estimate is immediate.

References

[1] M. Hintermuller, K. Papafitsoros, and C. N. Rautenberg, *Variable step mollifiers and applications*, Integral Equations and Operator Theory 92 (2020), article 53; arXiv:1910.02751.